CODING

Linear Regression – Apartment Rent Prediction

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = pd.read_csv("rent.csv")

X = data[['Area','Rooms','Location']]
y = data['Rent']

model = LinearRegression()
model.fit(X,y)

print("Intercept:",model.intercept_)
print("Coefficients:",model.coef_)

pred = model.predict([[1450,4,2]])
print("Predicted Rent =",pred[0])
```

2) Linear Classification – Spam Detection

```
import pandas as pd
from sklearn.linear_model import LogisticRegression

data = pd.read_csv("spam.csv")

X = data[['Offer','Free','Money','Winner']]
y = data['Spam']

model = LogisticRegression()
model.fit(X,y)

prediction = model.predict([[4,2,1,1]])

if prediction[0]==1:
   print("Spam Email")
else:
   print("Non Spam Email")
```

3) Naive Bayes – Flu Prediction

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB


data = pd.read_csv("flu.csv")


X = data[['Fever','Cough','Headache']]

y = data['Flu']


model = GaussianNB()

model.fit(X,y)


prediction = model.predict([[1,1,1]])


print("Prediction =",prediction[0])
```